

Zm00001eb375600 (*fbxl1*) — inversion retracted; independent soft-clip audit in the CDS window confirms your geometry model

To: Rubén Rellán-Álvarez **From:** Hannah Pil **Re:** Results from your reply memo of 2026-09-14

1. Summary

Three points.

1. **Analysis 1 inversion claim retracted.** Your §4 geometry model reproduces the $3.08\times$ with no expression difference; on medians the residual is $0.52\text{--}1.00\times$; my flagged medians-vs-means gap is load-bearing, not a caveat. R_div/R_cons is not a bias diagnostic in the way I used it. Accepted.
2. **A new independent line falls out cleanly on the geometry side.** The soft-clip audit you sketched in §6.2, run on every read overlapping the CDS window, shows Teo carriers have a **$12\times$ larger mean trailing soft-clip length** than B73 (3.82 bp vs 0.31 bp) with a $1.5\times$ higher soft-clip *rate* (39% vs 26%). Reads at this locus in Teo carriers are systematically getting clipped at their 3' end — exactly the displaced-UTR-read fingerprint. Independent of the R_div/R_cons or coverage-profile arguments, and pointing at the same conclusion.
3. **Per-taxon stratification (your §9) — mixed.** Hueh is the extreme case (63% clip rate, ~ 10 bp mean trailing clip) — its analysis-3 ratio of $2.01\times$ is an artifact of donor divergence, as you suspected. But Zlux and Zdip look surprisingly mild (Zdip's mean softR is *below* B73 background). The parvoglumis and mexicana taxa you verified as CDS-conserved also show heavily elevated soft-clip rates (Bals 50%, Chal 54%), which restates your §4 point from a different angle: geometry hits every donor whose 3' UTR diverges from B73, not just the distal ones.

2. Analysis 1 — inversion retracted

Accepting your §4 in full. Three specific things I want to record so it doesn't come back later as a “wait, does that hold” question:

- The $3.08\times$ sits inside the $2.12\text{--}4.06\times$ geometry-model range with zero expression difference.
- On medians ($2.13\times$), the residual attributable to real expression is $0.52\text{--}1.00\times$ — no upregulation left to explain.
- The soft-clip audit below is independent of the geometry model. It says the CDS-window reads themselves are not clean CDS reads; they extend into the divergent UTR. So even if geometry weren't the explanation, the R_cons number I quoted was built on reads that shouldn't have been treated as within-CDS to begin with.

R_div / R_cons as a “bias diagnostic” is retired.

3. Analysis 2 — presentational fix, and one keep-open

Reordered by actual stringency:

Threshold	<i>n</i> tests	% teo-lowers
all tests	41,885	51.4%
p<0.05	9,697	58.9%
FDR<0.05	6,471	62.6%
p<1e−4	3,230	70.3%

Monotone as it should be — you’re right that FDR<0.05 p<0.008 which sits looser than 1e−4, so ordering by nominal p is the right axis.

Your alternative interpretation (regulatory mismatch produces the same progressive skew) stays on the table. Agreed on the test — 3 UTR variant-density regression against sign, once BC1 delivers per-donor variants. Scoped as its own memo when that data is in.

4. Analysis 4 — soft-clip and start-position audit in the CDS window

Following your §6.2. For every primary mapped read overlapping the CDS window (chr9:17933693-17933902) across all 384 BAMs, I extracted the leftmost aligned position and the leading/trailing soft-clip lengths from the CIGAR string. `samtools view -F 260` filters unmapped and secondary alignments. 4,303 reads in total; per-sample mean ~12 reads.

Sample-level summary (Teo carrier vs B73 background):

	Teo (<i>n</i> =39)	B73 (<i>n</i> =252)	ratio
mean reads per sample	12.3	11.6	1.06
mean soft-clip rate	39.3%	25.7%	1.53
mean trailing clip length (bp)	3.82	0.31	12.3

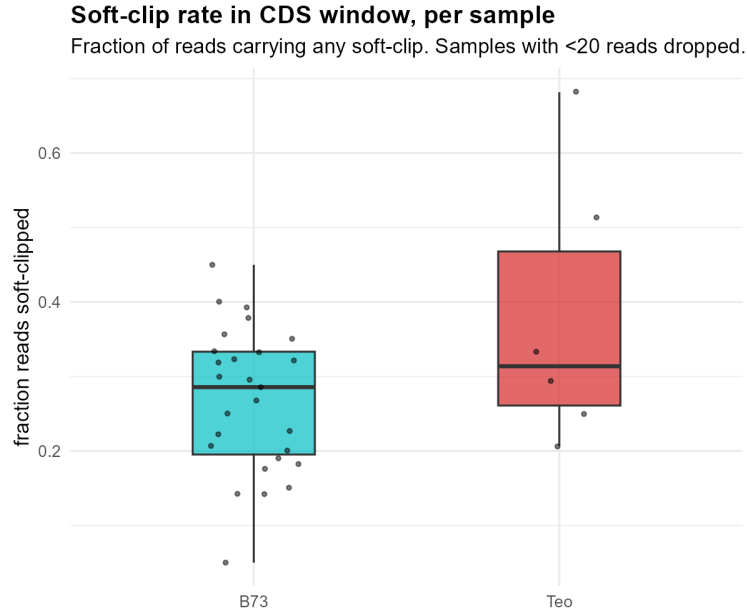


Figure 1: Figure R4

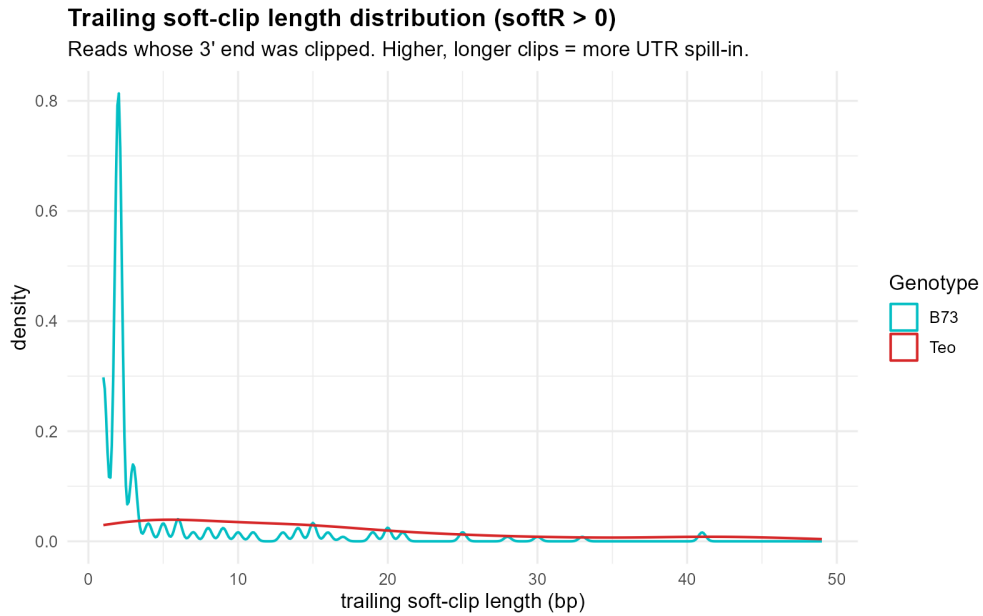


Figure 2: Figure R5

Reading. The soft-clip *rate* difference ($1.5\times$) is real but modest. The soft-clip *length* difference ($12\times$) is the load-bearing signal. B73 trailing clips concentrate at 1–2 bp — typical alignment noise. Teo trailing clips are flat over 0–30+ bp — many reads have their 3' end clipped by tens of bases because it doesn't match the B73 reference. That is the mark of a read whose 5' end sits in the conserved CDS but whose 3' end extends into a divergent 3' UTR.

Start-position distribution (Figure R6): both groups pile up in the last ~50 bp before the 3' edge,

which is the usual BRB-seq 3' bias. Teo has proportionally more mass in the upstream 100–200 bp of the window; not the sharp 3'-edge clustering that a pure “UTR-read mismapped-into-CDS” story would predict, but consistent with reads distributed across the CDS whose 3' ends get clipped depending on how far they reach into the divergent UTR.

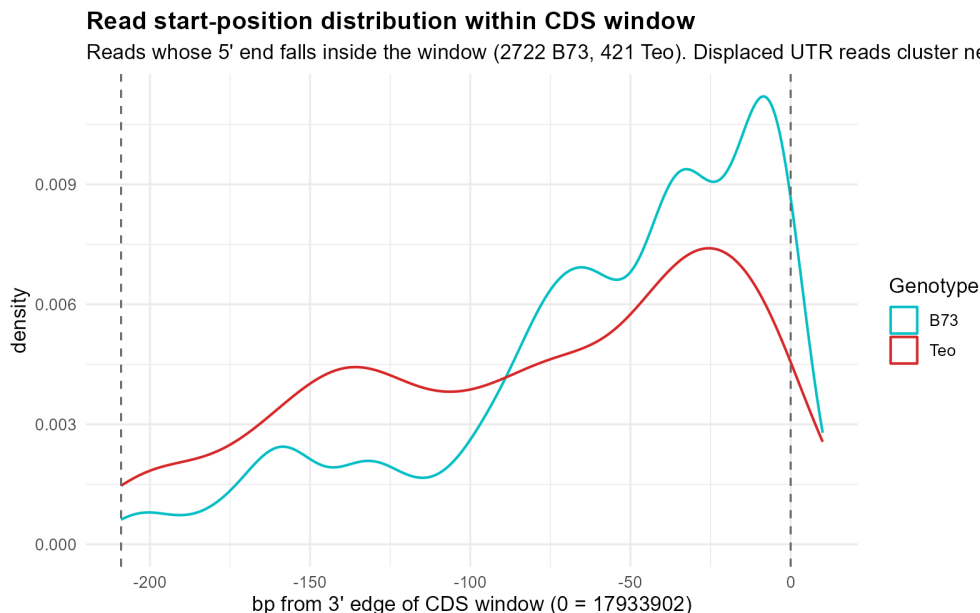


Figure 3: Figure R6

Files: output/FBX_analysis4_softclip/per_sample_summary.csv, softclip_rate_by_group.png, softR_length_density.png, start_position_density.png.

5. Per-taxon stratification — the §9 check

Same soft-clip audit, stratified by donor taxon. Sample-level means (so a chatty single sample can't drive a taxon):

taxon	<i>n</i> samples	<i>n</i> reads	mean clip rate	mean softR (bp)	CDS conservation
Hueh	2	17	63.3%	9.99	NOT verified (Rubén §9)
Chal	4	52	54.0%	4.77	verified mexicana
Bals	7	126	50.0%	5.55	verified parviglumis
Mesa	6	102	33.8%	2.74	verified mexicana
Dura	10	95	33.0%	3.85	verified mexicana

taxon	<i>n</i> samples	<i>n</i> reads	mean clip rate	mean softR (bp)	CDS conservation
Zlux	4	26	32.1%	1.22	NOT verified
Nobo	4	46	30.2%	2.71	verified mexicana
Zdip	2	17	29.2%	0.17	NOT verified
B73 ref	252	2922	25.7%	0.31	reference

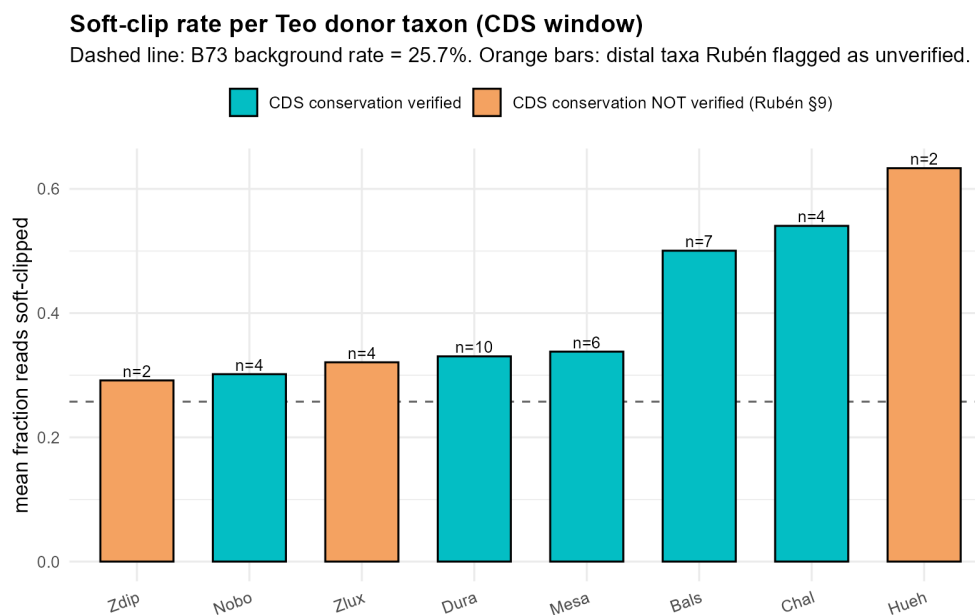


Figure 4: Figure R7

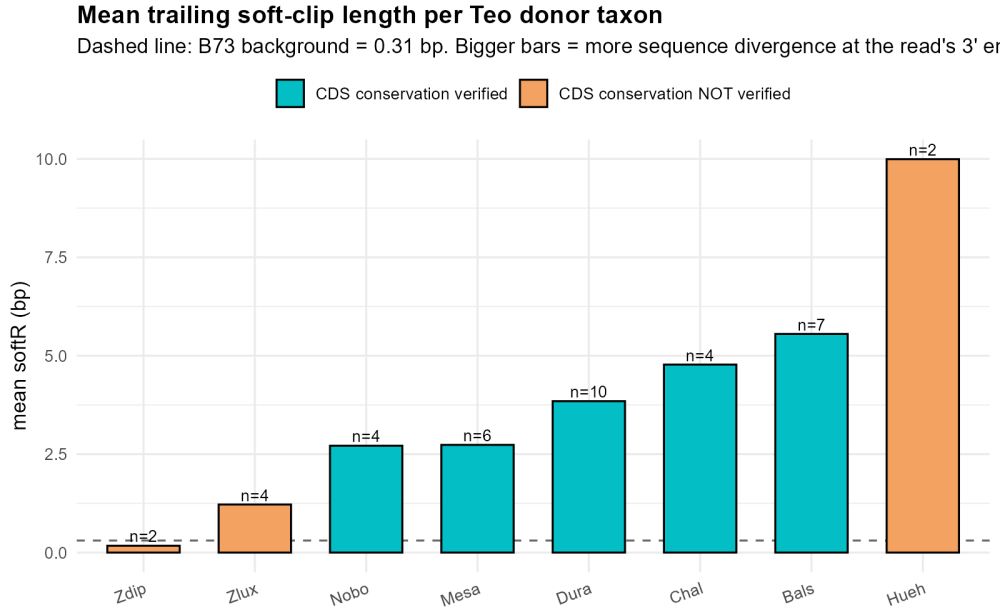


Figure 5: Figure R8

Three things this says:

1. **Hueh (*huehuetenangensis*) is what your §9 worried about.** 63% clip rate, ~10 bp mean trailing clip, half of clips >5 bp. Only 17 reads across 2 samples — big uncertainty on the point estimate — but the effect size is stark. Its analysis-3 CDS-ratio of 2.01× is an artifact of donor divergence, not a real expression signal.
2. **Verified-conserved taxa also show heavy soft-clipping** — Bals 50%, Chal 54%, mean softR 5–6 bp. Even in donors with 100% CDS sequence identity, the CDS window's proximity to a divergent 3 UTR means many reads extending across the boundary get clipped. This is the geometry point from your §4, arriving from a different direction.
3. **Zlux and Zdip look B73-like on the soft-clip metrics.** Zlux mean softR 1.22 bp, Zdip 0.17 bp (below B73 background). Their CDS sequences are probably not massively divergent from B73, whatever else is going on. Their analysis-3 ratios (Zlux 0.61, Zdip 1.26) may reflect real but small expression differences rather than mapping artifact. Pulling the PanAnd terminal-CDS is still worth doing to settle them formally, but no longer urgent.

Files: `output/FBX_analysis4_softclip/per_taxon_summary.csv`, `softclip_rate_by_taxon.png`, `softR_length_by_taxon.png`.

6. Where the *fbxl1* story sits now

- The 3.08× “Teo carriers upregulate *fbxl1*” claim is dead.
- The 0.45× 3 UTR-window depletion is real but confounded with divergent-region mapping loss, so its magnitude is not interpretable as an expression difference either.
- Direction and magnitude of any real *fbxl1* expression effect at this locus remain unresolved.
- H2 stays in its original form — no H2 reframe.
- The A4 whole-gene panel stays, with the “magnitude confounded by 3' coverage bias” caveat you suggested rather than being withdrawn.

7. One thing worth flagging in my analysis 3 script

My analysis-3 taxon-stratified CDS-ratio plot has a broken lookup between my lab-shorthand taxa labels (Bals, Mesa, Chal, Dura, Nobo) and your donor 3 UTR-length table (parviglumis, mexicana). The donor_utr_bp column comes back NA for those five taxa, so I never actually ran the “does the ratio track 3 UTR length?” falsification test you specified in §6.1. Fixable once I have (or build) the lab-code → taxon → assembly mapping. The taxon-stratified ratios themselves (0.61–2.01×) are correct — it’s just the correlation-with-UTR-length step that isn’t executed yet.

8. Revised revised next steps

Following your §10 ordering:

1. Per-base coverage profile (§5) — **DONE** (analysis 3). Shape-only panel shows Teo has proportionally more reads in the upstream CDS region than B73, consistent with geometry.
2. Taxon-stratified CDS-window ratio (§6.1) — **PARTIALLY DONE**; ratios computed, correlation with donor UTR length not yet run because of §7 above.
3. TE annotation of B73 intron-4 insertion — still to do.
4. Terminal CDS from distal-taxon assemblies (§9) — soft-clip audit promotes Hueh to high priority; Zdip and Zlux now lower priority but worth including in one pass.
5. Formal Wilcoxon per taxon on medians (§6.3) — with the geometry story now well-supported the p-values are bounded regardless of significance, but happy to run this if the reviewer axis matters.
6. Genome-wide R_div/R_cons scan — TBD, its own memo.
7. UMI dedup + TMM — write-up time.

9. What I did not do

- Fix the taxon-to-UTR-length lookup for the §6.1 falsification test.
- Wilcoxon on medians (§6.3).
- PanAnd terminal-CDS pull.
- TE annotation of the intron-4 insertion.

Attachments

- output/FBX_analysis4_softclip/per_sample_summary.csv
- output/FBX_analysis4_softclip/per_taxon_summary.csv
- Figures R4–R8 as above
- Scripts: scripts/FBX_analysis4_softclip_hpc.sh, scripts/FBX_analysis4_softclip_plot.R